

Definitions

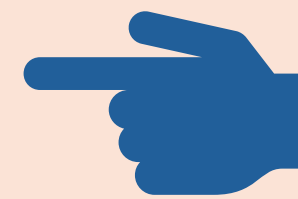
- **Closed-Form Solutions:** finite combos of standard operations (+, −, ×, ÷, √) and functions (exp, sin, etc.)
- **SymPy:** an open-source **computer-algebra package** for Python



- **The “Right” Way:** Galois (“gal-WAH”) theory, transcendental number theory, algebraic geometry.
- **The “Wrong” Way:** sympy.solve(...) *#NotAMathematician*

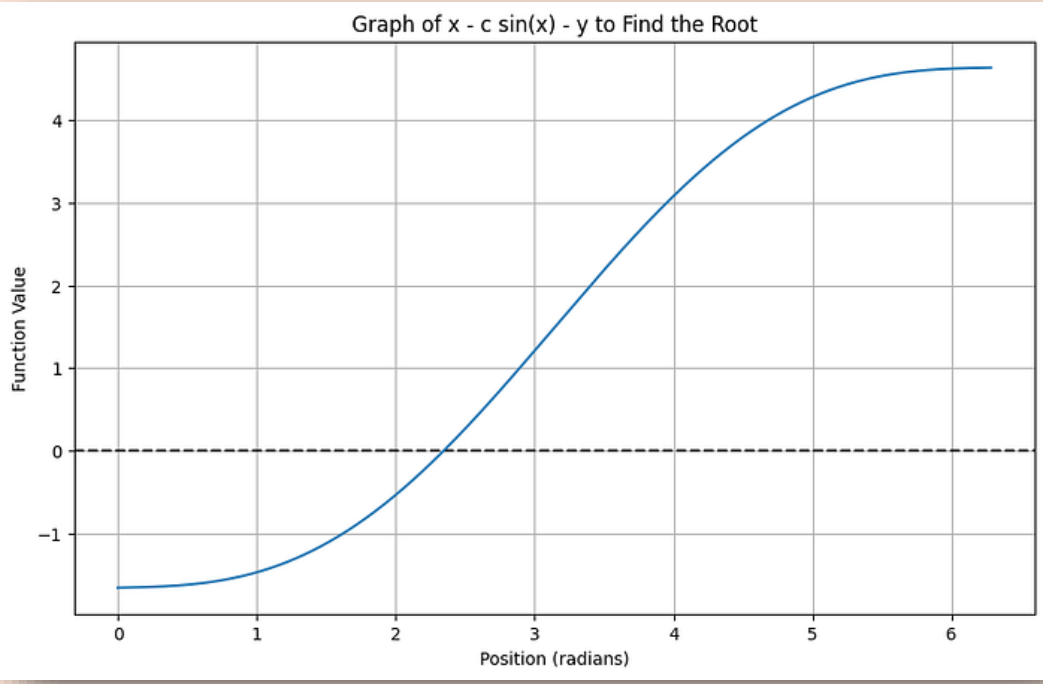
```
import sympy as sym
from sympy import pi, sin, Rational
from sympy.abc import x

equation = x - 0.967 * sin(x) - Rational(10, 19) * pi
try:
    solution = sym.solve(equation, x)
except Exception as e:
    print(f"Error: {e}")
```



NotImplementedError: multiple generators [x, sin(x)]
No algorithms are implemented to solve equation x - 967*sin(x)/1000 - 10*pi/19

- **The “Good Enough” Way:** sympy.nsolve(...)



```
position = sym.nsolve(equation, 2, 1.0)
print(f"After 20 Earth years, the comet will travel {position:.4f} radians ((position * 180 / pi).2f)° along its orbit.")
```

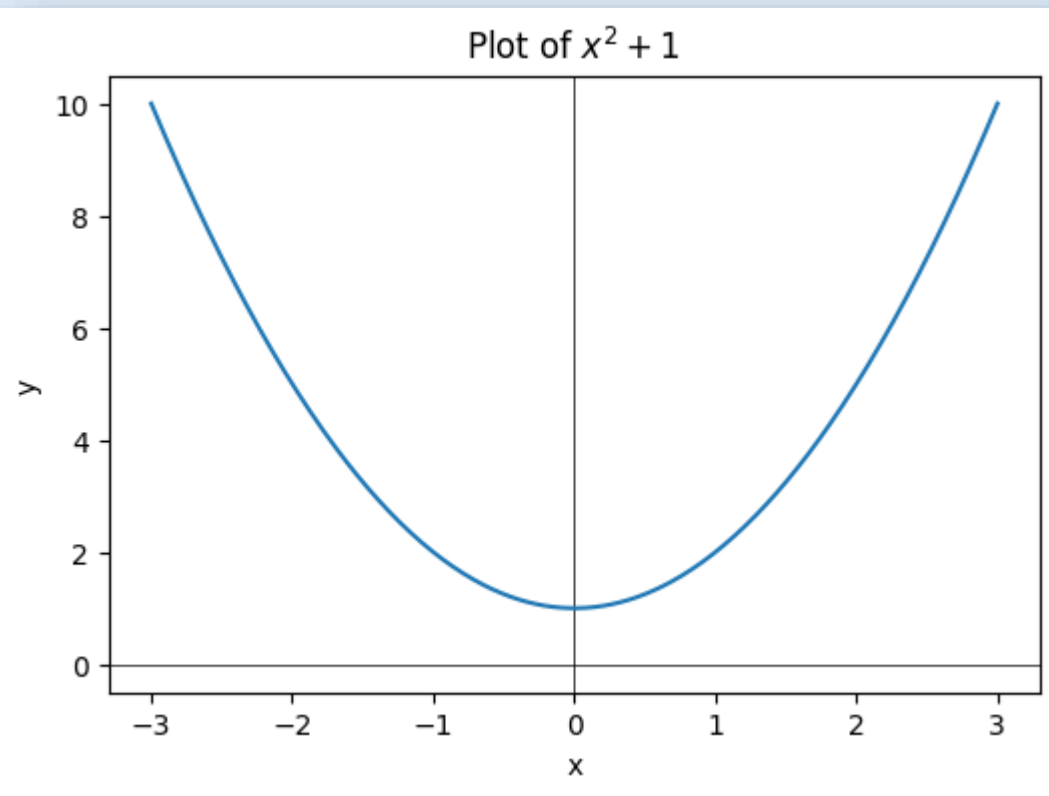
After 20 Earth years, the comet will travel 2.3449 radians (134.35°) along its orbit.

Equation	Closed-Form (CF) Solution?	CF Solution(s)	A Numeric
$x - 0.967 \sin(x) - \frac{10\pi}{19} = 0$	No? (NotImplementedError)		2.3449

Polynomials

Equation	Closed-Form (CF) Solution?	CF Solution(s)	A Numeric
$x^2 - x - 1 = 0$	Yes. Elementary	$\{\frac{1}{2} - \frac{\sqrt{5}}{2}, \frac{1}{2} + \frac{\sqrt{5}}{2}\}$	-0.6180
$2x^3 + x + 1 = 0$	Yes. Elementary	$\{\frac{1}{2(-\frac{1}{2} - \frac{\sqrt{3}i}{2})\sqrt{\frac{1}{12} + \frac{13\sqrt{3}i}{4}}}, \dots\}$	0.2949 + 0.8723i
$x^5 + 1 = 0$	Yes. Elementary	$\{-1, \frac{1}{4} + \frac{\sqrt{5}}{4} + i\sqrt{\frac{5}{8} - \frac{\sqrt{5}}{8}}, \dots\}$	-1.0000
$x^5 - x - 1 = 0$	No? (CRootOf)		1.1673

- ≤ degree 4, Closed Form
- May involve i, **complex** numbers



Why can we solve some equations easily, while others seem impossible?

Also: why is this knowledge hidden from us?

Challenge: Solve for x

- $y = x^2$ ✓
- $y = x - c \sin(x)$ ✗

Trick: “Cheat” with SymPy



A Python Refereeing an Italian Renaissance Mathematics Duel

Explore Solvable and Unsolvable Equations with SymPy

by Carl M. Kadie,
FaST-LMM Open-Source Genomics Project
<https://medium.com/@carlmkadie>



- Article
- Python Notebook
- Source Code
- <https://bit.ly/EqSolve>

Exp, Log and x

Equation	Closed-Form (CF) Solution?	CF Solution(s)	A Numeric
$xe^x = 0$	Yes. Elementary	0	0.0000
$xe^x + \frac{1}{10} = 0$	Yes, with W, Exp, Log	$\{W(-\frac{1}{10}), W_{-1}(-\frac{1}{10})\}$	-0.1118
$x + e^x = 0$	Yes, with W, Exp, Log	$-W(1)$	-0.5671
$x \log(x) + 1 = 0$	Yes, with W, Exp, Log	$e^{W(-1)}$	0.1684 + 0.7078i
$x + \log(x) = 0$	Yes, with W, Exp, Log	$W(1)$	0.5671
$x^2 + \log(x) = 0$	Yes, with W, Exp, Log	$e^{\frac{W(2)}{2}}$	0.6529

- Allow **Lambert’s W** function? Yes!

Equation	Closed-Form (CF) Solution?	CF Solution(s)	A Numeric
$e^x \log(x) + 1 = 0$	No? (NotImplementedError)		0.5671
$e^x + \log(x) = 0$	No? (NotImplementedError)		0.2699

Trigonometric

- Simple, yes
- Commensurate Frequency, depends
- Non-Commensurate, usually not

Trigonometric and x, Exp, Log

- Mostly nope

Equation	Closed-Form (CF) Solution?	CF Solution(s)	A Numeric
$\sin(x) \sin(x+1) + 1 = 0$	Yes. Elementary	$\{-2 \operatorname{atan}\left(\frac{\sin(\frac{1}{2})\sqrt{1+\sin^2(\frac{1}{2})}+2\sin^2(\frac{1}{2})-2\sqrt{\sin^2(\frac{1}{2})+2\sin(\frac{1}{2})+\sqrt{\sin^2(\frac{1}{2})+2}}}{\sin^2(\frac{1}{2})-1}\right), \dots\}$	-0.5000 + -0.7921i
$\sin(3x) \sin(x+1) + 1 = 0$	Timeout		ValueError
$\sin(\sqrt{3}x) \sin(x+1) + 1 = 0$	No? (NotImplementedError)		ValueError
$x + \sin(x) = 0$	No? (NotImplementedError)		0.0000

Takeaways

- **Nopes**
 - 5th degree polynomial
 - exp or log with x (Yep, with W)
 - exp and log together
 - Just Trig: see above
 - Trig and x/exp/log
- **Surprises**
 - Allow W
 - SymPy useful & fun
- Another project: *Newtonian physics isn’t deterministic*
- Numerical when needed

